Enrollment No.: _____

Darshan UNIVERSITY
योगः कर्मसु कौशलम्

Darshan Institute of Computer Application
BCA | Sem-3 | Winter-2023

Course Code	:	2104CS301	**Date**	:	04-11-2023
Course Name	:	Computer Network	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Compare star and bus topologies with figure.	**4**
	(B)	Define computer network with its characteristics.	**3**
		OR	
		Write short note on types of computer network.	
	(C)	Explain client server architecture in brief.	**7**
		OR	
		Define computer network. What are the applications of computer network? List advantages and disadvantages of computer network.	
Q.2	**(A)**	Explain application layer and list protocols working on it.	**4**
	(B)	List all functions of network layer and explain any one in brief.	**3**
		OR	
		Draw OSI reference Model.	
	(C)	Describe and draw TCP/IP Model.	**7**
		OR	
		Discuss all functions of data link layer and transport layer and explain any one function of both the layer in brief.	
Q.3	**(A)**	Write advantages and disadvantages of radio waves.	**4**
	(B)	Discuss the following terms: i) transmission media ii) guided media iii) attenuation	**3**
		OR	
		Explain coaxial cable with proper figure.	
	(C)	Describe micro wave.	**7**
		OR	
		Discuss fiber optic cable and twisted pair cable with suitable diagram.	

Q.4	**(A)**	Compare switch and router.	**4**
	(B)	Explain the Domain Name System.	**3**
		OR	
		Write a short note on LAN card.	
	(C)	Write functionality of router in networking.	**7**
		OR	
		Define networking device. List the devices used in networking. List types of switches. Write number of port switches available in market.	
Q.5	**(A)**	Slove the following questions: i) Write 192.168.90.64 into binary format. ii) Write 10110111 11101000 10111100 10000000 into decimal format.	**4**
	(B)	Write a short note on SMTP.	**3**
		OR	
		Write short note on DNS.	
	(C)	Subnet the IP address 216.21.5.0 into 30 hosts in each subnet.	**7**
		OR	
		A host has been assigned an IP address 192.168.17.9. Calculate the class, the number of addresses in the block, the first address, and the last address.	

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